

Summary and Highlights

What is the main purpose of Apache Cassandra?

The main purpose of **Apache Cassandra** is to provide a highly available, scalable, and fault-tolerant database solution for applications that require continuous access to data. Key features include:

- **High Availability:** Ensures that data is always accessible, even in the event of hardware failures.
- **Scalability:** Can handle large volumes of data and traffic by distributing data across multiple nodes and data centers.
- **Fault Tolerance:** Automatically replicates data across different nodes to prevent data loss.
- **Decentralized Architecture:** Eliminates single points of failure, allowing for seamless data distribution and management.

These characteristics make Cassandra particularly suitable for applications that demand reliability and performance in handling large datasets.

This lesson provides an overview of Apache Cassandra, a NoSQL database designed for high availability and scalability.

Apache Cassandra Overview

- It is an open-source, distributed, and decentralized database that is fault-tolerant and tunable.
- Best suited for applications requiring constant availability, with data distributed across one or more data center clusters.

Data Model and Structure

- Data is organized in tables within keyspaces, with clustering keys determining the order of data within partitions.
- Dynamic table partitions can grow as more entries are added, allowing for flexible data management.

Cassandra Query Language (CQL)

- CQL is the primary language for interacting with Cassandra clusters.
- Queries can be executed programmatically using a licensed client driver or through the Python-based CQL shell client.

Recap

- Apache Cassandra is an open source, distributed, decentralized, elastically scalable, highly available, fault-tolerant, tunable, and consistent database.
- Apache Cassandra is best used by applications that require a database that is always available.
- Data distribution and replication take place in one or more data center clusters.
- Cassandra stores data in tables that are grouped in keyspaces.
- A clustering key specifies the order in which the data is arranged inside the partition (ascending or descending).
- Dynamic table partitions grow dynamically with the number of entries.
- CQL is the primary language for communicating with Apache Cassandra clusters.
- CQL queries can be run programmatically using a licensed Cassandra client driver, or they can be run on the Python-based CQL shell client provided with Cassandra.

Keyspace	A keyspace in Cassandra is the highest-level organizational unit for data, similar to a database in traditional relational databases.
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Partition Key	The partition key is a component of the primary key and determines how data is distributed across nodes in a cluster.
Partitions	Partitions in Cassandra are the fundamental unit of data storage. Data is distributed across nodes and organized into partitions based on the partition key.
Peer-to-peer	The term peer-to-peer refers to the overall Cassandra architecture. In Cassandra, each node in the cluster has equal status and communicates directly with other nodes without relying on a central coordinator. If a primary node fails, another node automatically becomes the primary node.
Primary key	The primary key consists of one or more columns that uniquely identify rows in a table. The primary key includes a partition key and, optionally, clustering columns.
Replication	Replication involves creating and maintaining copies of data on multiple nodes to ensure data availability, reduce data loss, fault tolerance (improve system resilience), and provide read scalability.
Scalability	Scalability is the ability to add more nodes to the cluster to handle increased data and traffic.
Static table	A static table has a fixed set of columns for each row.
Table	A table is a collection of related data organized into rows and columns.